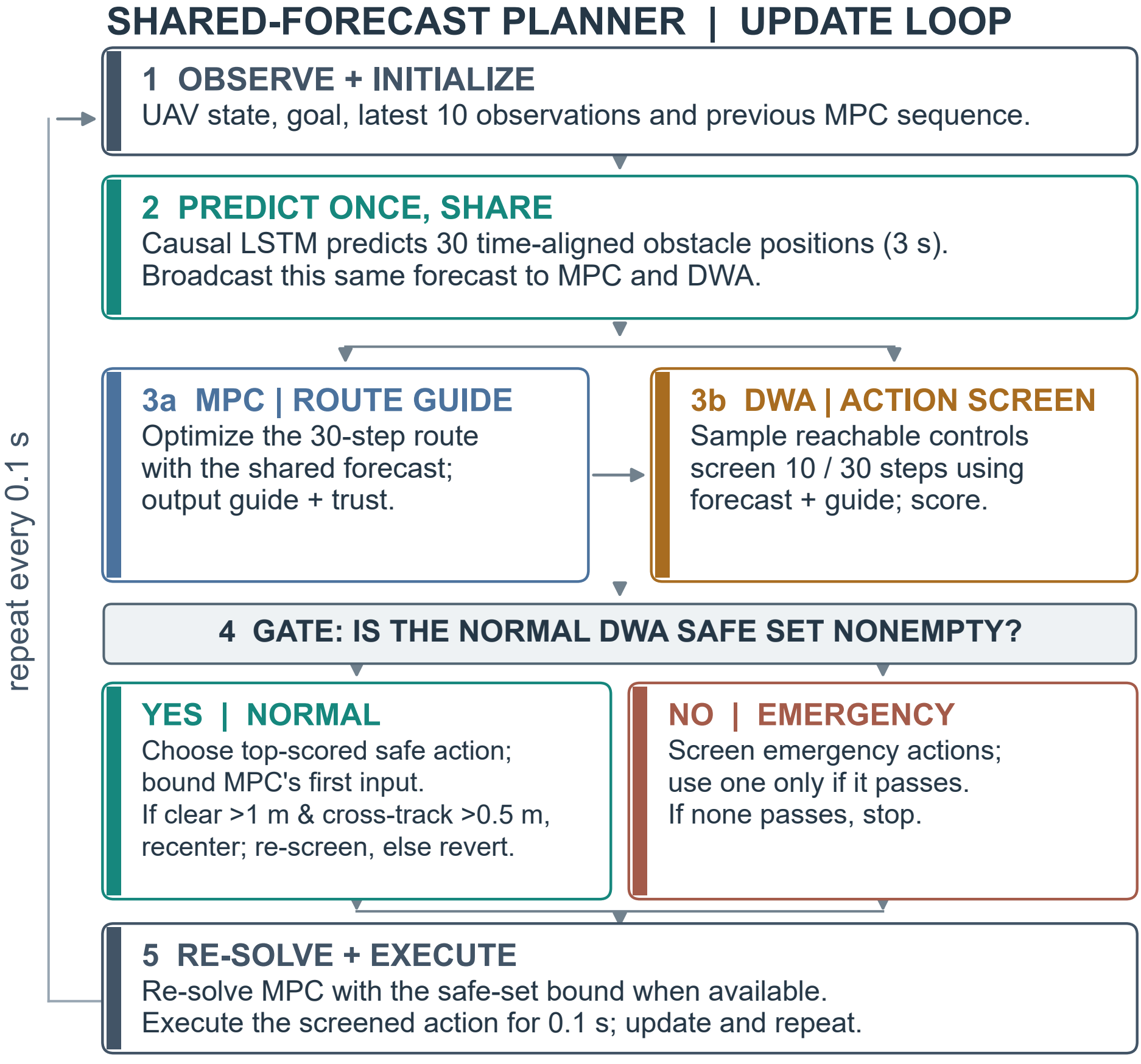

SHARED-FORECAST PLANNER | UPDATE LOOP
1 OBSERVE + INITIALIZE
UAV state, goal, latest 10 observations and previous MPC sequence.
2 PREDICT ONCE, SHARE
Causal LSTM predicts 30 time-aligned obstacle positions (3 s).
Broadcast this same forecast to MPC and DWA.
3a MPC | ROUTE GUIDE
Optimize the 30-step route
with the shared forecast;
output guide + trust.
3b DWA | ACTION SCREEN
Sample reachable controls
screen 10 / 30 steps using
forecast + guide; score.
4 GATE: IS THE NORMAL DWA SAFE SET NONEMPTY?
YES | NORMAL
Choose top-scored safe action;
bound MPC's first input.
If clear >1 m & cross-track >0.5 m,
recenter; re-screen, else revert.
NO | EMERGENCY
Screen emergency actions;
use one only if it passes.
If none passes, stop.
5 RE-SOLVE + EXECUTE
Re-solve MPC with the safe-set bound when available.
Execute the screened action for 0.1 s; update and repeat.
repeat every 0.1 s